Amazon Sales Data Analysis

1465 products • customer & product reviews • actionable insights

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Dataset Summary

Scope

Rows: 1465 • Columns: 15

Key Features

Customer demographics, product details, pricing, ratings, reviews

Data Quality

Missing: rating_count (2 nulls)

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Data Preparation

- Loaded with pandas; df.info() & describe()
- Imputed Review Rating nulls with category median
- Renamed columns to snake_case
- Engineered: age_group, purchase_frequency_days

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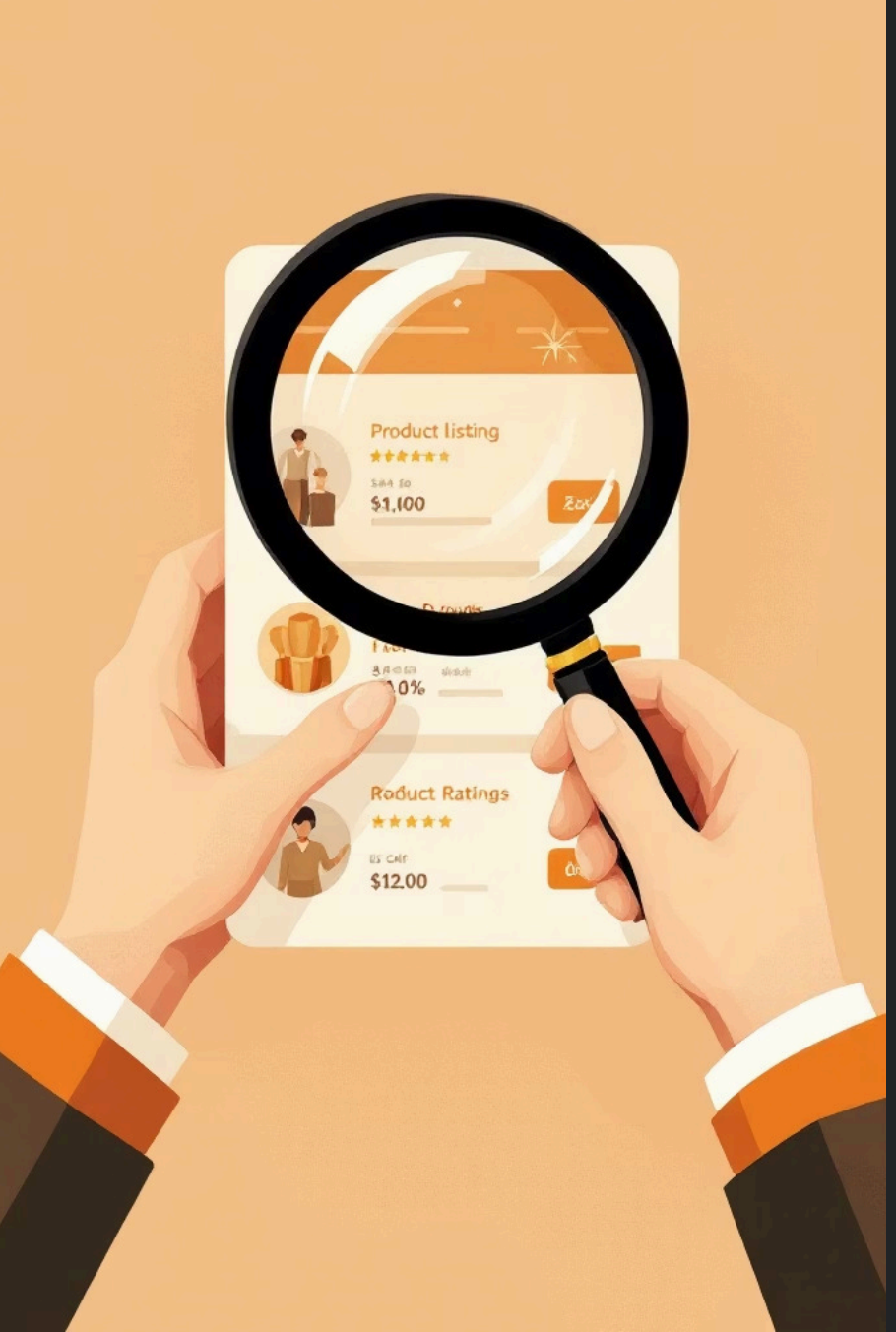

Exploratory Findings

Total Products

1465 products

Rating Avg

Mean rating = 4.0966

Discount Range

Min 0% — Max 94%

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Top-rated Products (examples)

REDTECH USB-C to Lightning Cable — 5.0

Syncwire LTG to USB Cable — 5.0

Amazon Basics Wireless Mouse — 5.0

Swiffer Instant Electric Faucet — 4.8

Oratech Coffee Frother — 4.8

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Power BI Dashboard

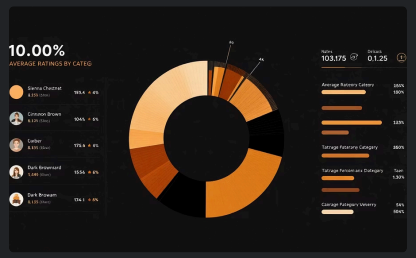
Interactive dashboard: key metrics + category filters + multiple charts

Top Metrics

Products: 1.465K • Avg discount: 48% • Avg price: 5.44K • Avg rating: 4.1

Price by Category

Bar chart: avg prices (0K–76K)



Rating by Category

Donut: avg rating range 3.8–4.6

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Category & Product Distribution

1194 unique customers • category-level counts via sunburst • product-level avg ratings

Insight: Basic USBs have more ratings

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Discount Insights

High Discounts

Max discount 94% — opportunity for targeted promotions

Price Variation

Avg price ≈ 5.44K — wide range across categories

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Boost Discounts

Customer Loyalty

Targeted Marketing

Sequence of recommended actions to increase revenue and retention

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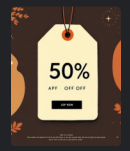
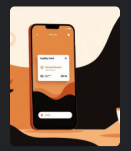
Business Recommendations



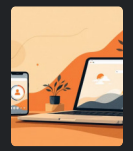
Boost Discounts

Promote exclusive benefits for customers



Customer Loyalty

Reward repeat buyers; subscription pathways



Targeted Marketing

Focus on high-revenue categories & express-shipping users

Product Positioning

Highlight top-rated & best-selling products in campaigns

Also: review discount policy to balance margin vs sales

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